

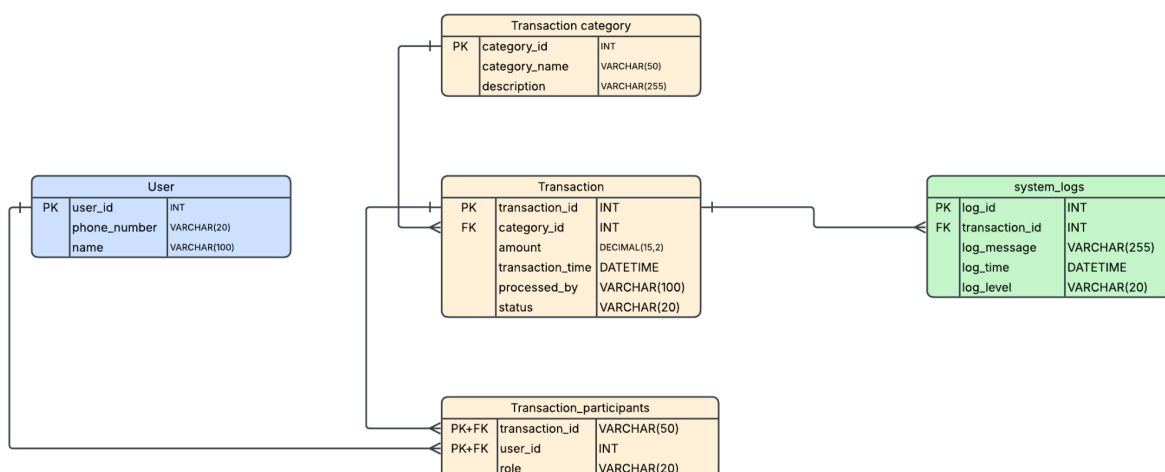
September 15th 2026
MoMo Analytics System
Database Design Document

EMS Systems: Erica Sheja, Ishimwe Samuel, Isingizwe Elohim Malaika
Week 2 – Database Design and Implementation

1. Overview

The MoMo Analytics System organizes and manages mobile money transaction data. The database stores information about users, transactions, transaction categories, transaction participants, and system logs. The design supports storing, organizing, and analyzing transaction information while maintaining relationships and data integrity across the system, ensuring users can trust its reliability.

2. Entity Relationship Diagram for the MoMo Analytics System(ERD)



3. Database Design Justification

Our MoMo Analytics System database is designed to organize transaction data while clearly representing the people, services, and activities involved in each transaction. The database contains five main entities: **Users**, **Transactions**, **Transaction Categories**, **Transaction Participants**, and **System Logs**.

The **Users** entity stores basic information about MoMo customers, such as their phone number and name. The **Transactions** entity records what happened, including the amount, time, processing service, status, and category. **Transaction Categories** describes the type of transaction, such as sending money, withdrawing money, purchasing airtime, or making a payment. This makes it easier to classify and analyze different transactions.

A key design decision we made was the **Transaction Participants** junction table. A transaction can involve multiple users, while a user can participate in many transactions. For example, when one user sends money to another, one user is recorded as the **sender** and the other as the **receiver**. The **role** attribute clearly identifies each participant's role, helping the audience trust the system's transparency and clarity.

The **processed_by** attribute identifies the system or service that processes the transaction, rather than the customer sending or receiving the money. **System Logs** records activities that occur while transactions are being processed. One transaction can generate multiple logs, such as processing or completion events.

The primary and foreign keys are essential for maintaining relationships and data integrity, while constraints help prevent invalid data entries across the system.

4. Data Dictionary

Users

- **user_id** — INT, Primary Key: Unique identifier for each user.
- **phone_number** — VARCHAR(20): The Phone number associated with the user's MoMo account.
- **name** — VARCHAR(100): Name of the user/customer.

Transactions

- **transaction_id** — INT, Primary Key: Unique identifier for each transaction.
- **amount** — DECIMAL(15,2): Money involved in the transaction.
- **transaction_time** — DATETIME: Date and time when the transaction occurred.
- **processed_by** — VARCHAR(100): System or service that processed the transaction.
- **status** — VARCHAR(20): Current processing status of the transaction.
- **category_id** — INT, Foreign Key: The category assigned to the transaction.

Transaction Categories

- **category_id** — INT, Primary Key: Unique identifier for each transaction category.
- **category_name** — VARCHAR(50): Name of the transaction category.
- **description** — VARCHAR(255): Explanation of what the category represents.

Transaction Participants

- **transaction_id** — INT, Primary Key & Foreign Key: Identifies the transaction in which the user participates.
- **user_id** — INT, Primary Key & Foreign Key: Identifies the user participating in the transaction.
- **role** — VARCHAR(20): Identifies the user's role, such as sender or receiver.
- Composite Primary Key: **transaction_id** + **user_id**.

System Logs

- **log_id** — INT, Primary Key: Unique identifier for each system log.
- **transaction_id** — INT, Foreign Key: Identifies the transaction associated with the log.
- **log_message** — VARCHAR(255): Message describing the system event.
- **log_level** — VARCHAR(20): Severity level of the log.
- **log_time** — DATETIME: Date and time when the log was created.

5. Sample SQL Queries

```
mysql> SHOW TABLES;
+-----+
| Tables_in_momo_analytics |
+-----+
| system_logs               |
| transaction_categories    |
| transaction_participants |
| transactions              |
| users                     |
+-----+
5 rows in set (0.008 sec)
```

```
mysql> SELECT * FROM users;
+-----+-----+-----+
| user_id | phone_number | name          |
+-----+-----+-----+
| 1       | 0788000001  | Jean Claude  |
| 2       | 0788000002  | Aline Mukamana |
| 3       | 0788000003  | Eric Niyonzima |
| 4       | 0788000004  | Grace Uwase  |
| 5       | 0788000005  | Samuel Habimana |
+-----+-----+-----+
5 rows in set (0.049 sec)
```

```
mysql> SELECT * FROM transactions;
+-----+-----+-----+-----+-----+-----+
| transaction_id | amount  | transaction_time | processed_by | status  | category_id |
+-----+-----+-----+-----+-----+-----+
| 1              | 5000.00 | 2026-09-10 08:30:00 | System      | SUCCESS | 1           |
| 2              | 12500.00 | 2026-09-10 09:15:00 | System      | SUCCESS | 2           |
| 3              | 2000.00  | 2026-09-10 10:00:00 | System      | SUCCESS | 3           |
| 4              | 30000.00 | 2026-09-10 11:45:00 | System      | PENDING | 4           |
| 5              | 15000.00 | 2026-09-10 13:20:00 | System      | FAILED  | 5           |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.009 sec)
```

6. Unique Rules / Constraints

- **Unique users:** Each user has a unique `user_id`, and phone numbers cannot be duplicated.
- **Valid transaction amounts:** Transaction amounts must be greater than zero.
- **Valid transaction status:** Transactions can only have the statuses `SUCCESS`, `PENDING`, or `FAILED`.
- **Valid participant roles:** A transaction participant can only have the role `SENDER` or `RECEIVER`.
- **Valid log levels:** System logs are restricted to `INFO`, `WARNING`, or `ERROR`.
- **Existing relationships:** Foreign keys ensure that transactions reference existing categories, participants reference existing users and transactions, and logs reference existing transactions.
- **No duplicate participation:** The combination of `transaction_id` and `user_id` prevents the same user from being recorded more than once for the same transaction.

Here are some selected constraints being enforced, such as:

- **An invalid transaction amount being rejected.**

```
mysql> INSERT INTO transactions
  -> (amount, transaction_time, processed_by, status, category_id)
  -> VALUES
  -> (0, '2026-09-15 14:00:00', 'System', 'SUCCESS', 1);
ERROR 3819 (HY000): Check constraint 'chk_transaction_amount' is violated.
```

- **An invalid transaction status being rejected.**

```
mysql> INSERT INTO transactions
  -> (amount, transaction_time, processed_by, status, category_id)
  -> VALUES
  -> (1000, '2026-09-15 14:00:00', 'System', 'INVALID', 1);
ERROR 3819 (HY000): Check constraint 'chk_transaction_status' is violated.
```

- **An invalid participant role being rejected.**

```
mysql> INSERT INTO transaction_participants
  -> (transaction_id, user_id, role)
  -> VALUES
  -> (1, 3, 'INVALID');
ERROR 3819 (HY000): Check constraint 'chk_participant_role' is violated.
```

- **An invalid log level being rejected.**

```
mysql> INSERT INTO system_logs
  -> (transaction_id, log_message, log_level, log_time)
  -> VALUES
  -> (1, 'Invalid log test', 'INVALID', '2026-09-15 14:00:00');
ERROR 3819 (HY000): Check constraint 'chk_log_level' is violated.
```